



NEUROSCIENCE

Unlocking the secrets of self-awareness

Long neglected by researchers, metacognition has made its way to the scientific spotlight

By **Christian C. Ruff**

For millennia, religious thinkers and philosophers have cited humanity's self-awareness—that is, our ability to think about our own mind and character—as being key to the uniqueness of our species. Carl Linnaeus's groundbreaking biological taxonomy (*I*) likewise characterized our genus by the words “Homo. Nosce te ipsum” (“Man. Know thyself”).

Given our long-standing interest in self-awareness, it is surprising how little science has traditionally had to say about it. What features of our brains enable us to think about ourselves? What are our strengths and weaknesses in this respect and how do they influence how we decide, learn, and interact? Can we train self-awareness, and how does this improve our performance? In the past three decades, however, research addressing such questions has been picking up speed. In *Know Thyself*, cognitive neuroscientist Stephen Fleming synthesizes this multifaceted research into an admirably coherent narrative and outlines how the resulting knowledge may be applied to solve societal problems.

Writing about self-awareness is challenging because concepts such as “self” and “awareness”—let alone the combination thereof—are hard to define. The book does not get lost in this epistemological Bermuda triangle but rather conceptualizes self-awareness as the set of mental and brain processes that keep track of our percepts, thoughts, and actions.

Not all of these metacognitive processes concern the self in a philosophical sense, Fleming notes, and not all of them need to be conscious. A helpful metaphor in the book compares the human brain to a flying plane that is largely guided by autopilot technology but can be flexibly controlled by the pilot whenever needed. For our behavior, the autopilot is the unconscious,

“implicit” part of metacognition, and the pilot is the “explicit” metacognition that we can consciously report.

Fleming begins by summarizing the psychology and neuroscience of these metacognitive processes. Implicit metacognition, he notes, is evident in many seemingly low-level brain processes, ranging from the sensory brain cells that signal the uncertainty associated with particular percepts, to brain cells that activate when we commit action errors (think: mistyping on a keyboard). All of these implicit signals can be read out in the service of explicit metacognition, when, for example, we need to judge our confidence in having



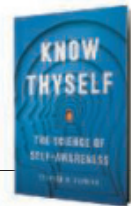
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chosen the right action. This latter ability depends on specific brain areas in the prefrontal cortex and is independent of the basic perceptual and motor abilities it serves to monitor.

Explicit metacognition, meanwhile, depends on our ability to think about the mental states of others—an ironic twist nicely summarized by the caption of a cartoon that appears in the book: “Of course I care about how you imagined I thought you perceived I wanted you to feel.”

It is eye-opening to realize how many fields of human endeavor depend not just on our skills and knowledge but also on our ability to estimate our competence.

**Know Thyself:
The Science
of Self-Awareness**
Stephen M. Fleming
Basic Books, 2021. 304 pp.



Obvious examples can be found in education, politics, the legal system, corporate decision-making and leadership, news and social media, and, indeed, any collaboration in which people pool their expertise. The book illustrates the role of metacognition in these diverse fields with elegant combinations of philosophical considerations, basic science findings, and more applied examples.

Fleming even ventures into the near future, sketching how artificial intelligence with superhuman computational abilities but no self-awareness may become disconnected from humanity at best and outright catastrophic at worst. Emerging ideas on how we may address this problem—for example, by endowing intelligent machines with coarse self-awareness or by ensuring that self-aware humans remain at the helm—only serve to prove how little we have appreciated our own prodigious metacognitive abilities.

In the end, the book makes a convincing case that self-awareness is a key feature of human existence and that our growing knowledge about it will be important for addressing many of our societal problems. One may quibble that the book somewhat understates this point, because it focuses on metacognition and does not cover our ability to monitor our emotions, another key aspect of self-awareness that has major implications for health and well-being. However, the literature on this topic is so diverse that doing it justice would likely require several additional volumes. As it stands, Fleming's book finally heaves metacognition into a long-deserved place in the scientific spotlight. ■

REFERENCES AND NOTES

1. C. Linnaeus, *Systema Naturae* (Haak, 1735).

10.1126/science.abh2030

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(unlock the phone, unlock the secrets/mysteries)

self-awareness n. 自我意识，自我认知

(good knowledge and judgment about yourself)

neglect v. & n. 忽略，忽视

metacognition n. 元认知 (knowledge and understanding of your own thinking)

(Metacognition is considered one of humans' most sophisticated cognitive capacities.)

make one's way to 设法到达

spotlight n. 聚光灯

(in the spotlight 备受关注的)

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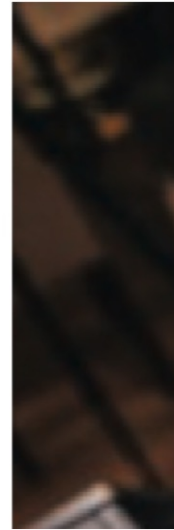
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(His latest movie is interesting, but not groundbreaking.)

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likewise adv. 同样地，也
(Nice to meet you. Likewise.)

genus n. 类，种，属

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Self-aware
well-learn

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metaphor

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prefrontal cortex

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(completely, directly or immediately instead of in stages)

(They bought their car outright. / You're telling an outright lie.)

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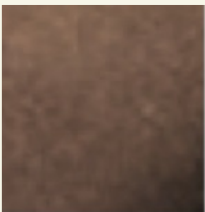
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READING

READING PASSAGE 1

You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.

The concept of intelligence

- A** Looked at in one way, everyone knows what intelligence is; looked at in another way, no one does. In other words, people all have unconscious notions – known as ‘implicit theories’ – of intelligence, but no one knows for certain what it actually is. This chapter addresses how people conceptualize intelligence, whatever it may actually be.

But why should we even care what people think intelligence is, as opposed only to valuing whatever it actually is? There are at least four reasons people’s conceptions of intelligence matter.

- B** First, implicit theories of intelligence drive the way in which people perceive and evaluate their own intelligence and that of others. To better understand the judgments people make about their own and others’ abilities, it is useful to learn about people’s implicit theories. For example, parents’ implicit theories of their children’s language development will determine at what ages they will be willing to make various corrections in their children’s speech. More generally, parents’ implicit theories of intelligence will determine at what ages they believe their children are ready to perform various cognitive tasks. Job interviewers will make hiring decisions on the basis of their implicit theories of intelligence. People will decide who to be friends with on the basis of such theories. In sum, knowledge about implicit theories of intelligence is important because this knowledge is so often used by people to make judgments in the course of their everyday lives.
- C** Second, the implicit theories of scientific investigators ultimately give rise to their explicit theories. Thus it is useful to find out what these implicit theories are. Implicit theories provide a framework that is useful in defining the general scope of a phenomenon – especially a not-well-understood phenomenon. These implicit theories can suggest what aspects of the phenomenon have been more or less attended to in previous investigations.
- D** Third, implicit theories can be useful when an investigator suspects that existing explicit theories are wrong or misleading. If an investigation of implicit theories reveals little correspondence between the extant implicit and explicit theories, the implicit theories may be wrong. But the possibility also needs to be taken into account that the explicit theories are wrong and in need of correction or supplementation. For example, some implicit theories of intelligence suggest the need for expansion of some of our explicit theories of the construct.

- E** Finally, understanding implicit theories of intelligence can help elucidate developmental and cross-cultural differences. As mentioned earlier, people have expectations for intellectual performances that differ for children of different ages. How these expectations differ is in part a function of culture. For example, expectations for children who participate in Western-style schooling are almost certain to be different from those for children who do not participate in such schooling.
- F** I have suggested that there are three major implicit theories of how intelligence relates to society as a whole (Sternberg, 1997). These might be called Hamiltonian, Jeffersonian, and Jacksonian. These views are not based strictly, but rather, loosely, on the philosophies of Alexander Hamilton, Thomas Jefferson, and Andrew Jackson, three great statesmen in the history of the United States.
- G** The Hamiltonian view, which is similar to the Platonic view, is that people are born with different levels of intelligence and that those who are less intelligent need the good offices of the more intelligent to keep them in line, whether they are called government officials or, in Plato's term, philosopher-kings. Herrnstein and Murray (1994) seem to have shared this belief when they wrote about the emergence of a cognitive (high-IQ) elite, which eventually would have to take responsibility for the largely irresponsible masses of non-elite (low-IQ) people who cannot take care of themselves. Left to themselves, the unintelligent would create, as they always have created, a kind of chaos.
- H** The Jeffersonian view is that people should have equal opportunities, but they do not necessarily avail themselves equally of these opportunities and are not necessarily equally rewarded for their accomplishments. People are rewarded for what they accomplish, if given equal opportunity. Low achievers are not rewarded to the same extent as high achievers. In the Jeffersonian view, the goal of education is not to favor or foster an elite, as in the Hamiltonian tradition, but rather to allow children the opportunities to make full use of the skills they have. My own views are similar to these (Sternberg, 1997).
- I** The Jacksonian view is that all people are equal, not only as human beings but in terms of their competencies – that one person would serve as well as another in government or on a jury or in almost any position of responsibility. In this view of democracy, people are essentially intersubstitutable except for specialized skills, all of which can be learned. In this view, we do not need or want any institutions that might lead to favoring one group over another.
- J** Implicit theories of intelligence and of the relationship of intelligence to society perhaps need to be considered more carefully than they have been because they often serve as underlying presuppositions for explicit theories and even experimental designs that are then taken as scientific contributions. Until scholars are able to discuss their implicit theories and thus their assumptions, they are likely to miss the point of what others are saying when discussing their explicit theories and their data.

READING PASSAGE 2

You should spend about 20 minutes on **Questions 14–26** which are based on Reading Passage 2 below.

Young children's sense of identity

- A** A sense of self develops in young children by degrees. The process can usefully be thought of in terms of the gradual emergence of two somewhat separate features: the *self as a subject*, and the *self as an object*. William James introduced the distinction in 1892, and contemporaries of his, such as Charles Cooley, added to the developing debate. Ever since then psychologists have continued building on the theory.
- B** According to James, a child's first step on the road to self-understanding can be seen as the recognition that he or she exists. This is an aspect of the self that he labelled 'self-as-subject', and he gave it various elements. These included an awareness of one's own agency (i.e. one's power to act), and an awareness of one's distinctiveness from other people. These features gradually emerge as infants explore their world and interact with caregivers. Cooley (1902) suggested that a sense of the self-as-subject was primarily concerned with being able to exercise power. He proposed that the earliest examples of this are an infant's attempts to control physical objects, such as toys or his or her own limbs. This is followed by attempts to affect the behaviour of other people. For example, infants learn that when they cry or smile someone responds to them.
- C** Another powerful source of information for infants about the effects they can have on the world around them is provided when others mimic them. Many parents spend a lot of time, particularly in the early months, copying their infant's vocalizations and expressions. In addition, young children enjoy looking in mirrors, where the movements they can see are dependent upon their own movements. This is not to say that infants recognize the reflection as their *own* image (a later development). However, Lewis and Brooks-Gunn (1979) suggest that infants' developing understanding that the movements they see in the mirror are contingent on their own, leads to a growing awareness that they are distinct from other people. This is because they, and only they, can change the reflection in the mirror.
- D** This understanding that children gain of themselves as active agents continues to develop in their attempts to co-operate with others in play. Dunn (1988) points out that it is in such day-to-day relationships and interactions that the child's understanding of his- or herself emerges. Empirical investigations of the self-as-subject in young children are, however, rather scarce because of difficulties of communication: even if young infants can reflect on their experience, they certainly cannot express this aspect of the self directly.

- E Once children have acquired a certain level of self-awareness, they begin to place themselves in a whole series of categories, which together play such an important part in defining them uniquely as 'themselves'. This second step in the development of a full sense of self is what James called the 'self-as-object'. This has been seen by many to be the aspect of the self which is most influenced by social elements, since it is made up of social roles (such as student, brother, colleague) and characteristics which derive their meaning from comparison or interaction with other people (such as trustworthiness, shyness, sporting ability).
- F Cooley and other researchers suggested a close connection between a person's own understanding of their identity and other people's understanding of it. Cooley believed that people build up their sense of identity from the reactions of others to them, and from the view they believe others have of them. He called the self-as-object the 'looking-glass self', since people come to see themselves as they are reflected in others. Mead (1934) went even further, and saw the self and the social world as inextricably bound together: 'The self is essentially a social structure, and it arises in social experience ... it is impossible to conceive of a self arising outside of social experience.'
- G Lewis and Brooks-Gunn argued that an important developmental milestone is reached when children become able to recognize themselves visually without the support of seeing contingent movement. This recognition occurs around their second birthday. In one experiment, Lewis and Brooks-Gunn (1979) dabbed some red powder on the noses of children who were playing in front of a mirror, and then observed how often they touched their noses. The psychologists reasoned that if the children knew what they usually looked like, they would be surprised by the unusual red mark and would start touching it. On the other hand, they found that children of 15 to 18 months are generally not able to recognize themselves unless other cues such as movement are present.
- H Finally, perhaps the most graphic expressions of self-awareness in general can be seen in the displays of rage which are most common from 18 months to 3 years of age. In a longitudinal study of groups of three or four children, Bronson (1975) found that the intensity of the frustration and anger in their disagreements increased sharply between the ages of 1 and 2 years. Often, the children's disagreements involved a struggle over a toy that none of them had played with before or after the tug-of-war: the children seemed to be disputing ownership rather than wanting to play with it. Although it may be less marked in other societies, the link between the sense of 'self' and of 'ownership' is a notable feature of childhood in Western societies.

LETTERS

Edited by Jennifer Sills

Editorial Expression of Concern

On 27 March 2020, *Science* published the Research Article “Flux-induced topological superconductivity in full-shell nanowires” by S. Vaitiekėnas *et al.* (1). Pursuant to a reader request, the authors released additional data—archived at Zenodo (2)—taken in association with the project that led to their paper. After the release of the additional data, two readers expressed a joint concern that the tunneling spectroscopy data published in the original paper are not representative of the entirety of the data released in association with this project. While we await the outcome of a full investigation commenced by the authors’ academic institution (Niels Bohr Institute, University of Copenhagen), we are alerting our readers to this concern.

H. Holden Thorp
Editor-in-Chief

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Mitigate human-wildlife conflict in China

Since passing a wildlife protection law in 1989, China has made great strides in wildlife conservation (1). Protection in natural habitats, conservation in zoos, and release of captive wildlife back into nature have all facilitated the population growth of species such as the Siberian tiger (*Panthera tigris ssp. altaica*) and Amur leopard (*Panthera pardus orientalis*) (2). These animals, in search of additional food and space, enter areas developed by humans, leading to human-wildlife conflicts (3, 4). From the late 1990s through 2010, there were more than 6000 cases of compensation for wildlife damage nationwide (5). Yet, local populations and governments remain unprepared to deal with wildlife in populated areas.

Since March, 2020, a herd of wild Asian elephants (*Elephas maximus*)—an endangered species that originally inhabited the Xishuangbanna National Nature Reserve

in Yunnan’s southernmost prefecture—has been on the move, causing great damage to human interests and attracting international attention (6). Other large endangered wildlife, such as the gray wolf (*Canis lupus*), the Tibetan brown bear (*Ursus arctos pruinosus*), and the snow leopard (*Panthera uncia*), have also frequently left their established habitats and appeared in China’s urban and rural areas in recent years (7–9). In response, local governments have enacted temporary emergency plans and evacuations and, in some cases, rounded up or hunted the animals (10).

These measures threaten the safety and property of local residents and cause irreparable damage to wildlife (11). Instead, China should build on previous successes in ecological restoration and environmental governance. For example, the 13th Five-Year Plan included an ecological monitoring system based on community participation and an intelligent early warning platform that effectively protected 90% of the country’s plants and 85% of its wild animals (12).

To protect both humans and wildlife, local governments should invest in additional wildlife monitoring and biological diversity research. In addition, the National Forestry and Grassland Administration should provide the public with information about how to behave when confronted with wild animals, emphasizing the need to avoid personal injury and minimize economic loss while also keeping the animals

safe. News organizations should curb their tendency to sensationalize wildlife news and instead disseminate accurate information about endangered species and conflict prevention. To address wildlife encounters, the government should establish consistent emergency plans that include early evacuation when appropriate and the prompt notification of wildlife professionals, who can decide how to proceed based on the condition of the wild animals. Finally, a compensation mechanism should be put in place that incentivizes animal protection by providing more support for damages to property if the animal who caused them was not harmed.

Duo Yin, Zhenjie Yuan, Jie Li, Hong Zhu*

School of Geography and Remote Sensing, Guangzhou University, Higher Education Mega Center, Guangzhou, 510006, China.

*Corresponding author.

Email zhuhong@gzhu.edu.cn

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PHOTO: XINHUA/GETTY IMAGES

of Medicine, says both methods work well. “Both the DeepMind and Baker lab advances are phenomenal and will change how we can use protein structure predictions to advance biology,” she says. A DeepMind spokesperson wrote in an email, “It’s great to see examples such as this where the protein folding community is building on AlphaFold to work towards our shared goal of increasing our understanding of structural biology.”

But AlphaFold2 solved the structures of only single proteins, whereas RoseTTA-Fold has also predicted complexes, such as the structure of the immune molecule interleukin-12 latched onto its receptor. Many biological functions depend on protein-protein interactions, says Torsten Schwede, a computational structural biologist at the University of Basel. “The ability to handle protein-protein complexes directly from sequence information makes it extremely attractive for many questions in biomedical research.”

Baker concedes that, in general, AlphaFold2’s structures are more accurate. But Savvides says the Baker lab’s approach better captures “the essence and particularities of protein structure,” such as identifying strings of atoms sticking out of the sides of the protein—features key to interactions between proteins. Agard adds that Baker’s and Baek’s approach is faster and requires less computing power than DeepMind’s, which relied on Google’s massive servers. However, the DeepMind spokesperson wrote that its latest algorithm is more than 16 times as fast as the one it used at CASP in 2020. As a result, she wrote, “It’s not clear to us that the system being described is an advance in speed.”

Beginning on 1 June, Baker and Baek began to challenge their method by asking researchers to send in their most baffling protein sequences. Fifty-six head scratchers arrived in the first month, all of which have now predicted structures. Agard’s group sent in an amino acid sequence with no known similar proteins. Within hours, his group got a protein model back “that probably saved us a year of work,” Agard says. Now, he and his team know where to mutate the protein to test ideas about how it functions.

Because Baek’s and Baker’s group has released its computer code on the web, others can improve on it; the code has been downloaded 250 times since 1 July. “Many researchers will build their own structure prediction methods upon Baker’s work,” says Jinbo Xu, a computational structural biologist at the Toyota Technological Institute at Chicago. Moulton agrees: “When there’s a breakthrough like this, 2 years later, everyone is doing it as well if not better than before.” ■

NEUROSCIENCE

Brain signals ‘speak’ for person with paralysis

Algorithm creates words, sentences from neural activity

By **Kelly Servick**

A man unable to speak after a stroke has produced sentences through a system that reads electrical signals from speech production areas of his brain, researchers report this week. The approach has previously been used in nondisabled volunteers to reconstruct spoken or imagined sentences. But this first demonstration in a person who is paralyzed “tackles really the main issue that was left to be tackled—bringing this to the patients that really need it,” says Christian Herff, a computer scientist at Maastricht University who was not involved in the new work.

The participant had a stroke more than a decade ago that left him with anarthria—an inability to control the muscles involved in speech. Because his limbs are also paralyzed, he communicates by selecting letters on a screen using small movements of his head, producing roughly five words per minute. To enable faster, more natural communication, neurosurgeon Edward Chang of the University of California, San Francisco, tested an

approach that uses a computational model known as a deep-learning algorithm to interpret patterns of brain activity in the sensorimotor cortex, a brain region involved in producing speech (*Science*, 4 January 2019, p. 14). The approach has so far been tested in volunteers who have electrodes surgically implanted for nonresearch reasons such as to monitor epileptic seizures.

In the new study, Chang’s team temporarily removed a portion of the participant’s skull and laid a thin sheet of electrodes smaller than a credit card directly over his sensorimotor cortex. To “train” a computer algorithm to associate brain activity patterns with the onset of speech and with particular words, the team needed reliable information about what the man intended to say and when.

So the researchers repeatedly presented one of 50 words on a screen and asked the man to attempt to say it on cue. Once

the algorithm was trained with data from the individual word task, the man tried to read sentences built from the same set of 50 words, such as “Bring my glasses, please.” To improve the algorithm’s guesses, the researchers added a processing component called a natural language model, which uses common word sequences to predict the likely next word in a sentence. With that approach, the system only got about 25% of the words in a sentence wrong, they report this week in *The New England Journal of Medicine*. That’s “pretty impressive,” says Stephanie Riès-Cornou, a neuroscientist at San Diego State University. (The error rate for chance performance would be 92%.)

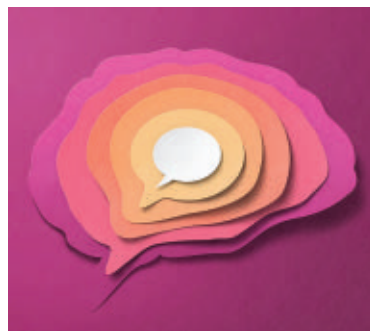
Because the brain reorganizes over time, it wasn’t clear that speech production areas would give interpretable signals after

more than 10 years of anarthria, notes Anne-Lise Giraud, a neuroscientist at the University of Geneva. The signals’ preservation “is surprising,” she says. And Herff says the team made a “gigantic” step by generating sentences as the man was attempting to speak rather than from previously recorded

brain data, as most studies have done.

With the new approach, the man could produce sentences at a rate of up to 18 words per minute, Chang says. That’s roughly comparable to the speed achieved with another brain-computer interface, described in *Nature* in May. That system decoded individual letters from activity in a brain area responsible for planning hand movements while a person who was paralyzed imagined handwriting. These speeds are still far from the 120 to 180 words per minute typical of conversational English, Riès-Cornou notes, but they far exceed what the participant can achieve with his head-controlled device.

The system isn’t ready for use in everyday life, Chang notes. Future improvements will include expanding its repertoire of words and making it wireless, so the user isn’t tethered to a computer roughly the size of a minifridge. ■



WHETHER on page, stage or screen, the story of human health and happiness is often presented as an inevitable arc between birth and death. William Shakespeare captured this best with his “seven ages of man” speech. We enter the world “mewling and puking” as an infant, pass through the awkwardness of childhood and adolescence into our physical and mental prime, before a slow decline.

Until recently, science appeared to confirm this view. For many abilities, we seemed to reach our peak well before midlife. But it is now becoming clear that this picture is far too simplistic. Childhood and adolescence may offer the most rapid periods of development, but our brains can change in positive ways throughout life, with some important cognitive skills continuing to improve into

our 50s, 60s and 70s. “The whole idea that the brain is fully mature at age 25 is a joke,” says Daniel Romer, a psychologist at the University of Pennsylvania.

Nor does our fitness simply rise, peak and fall in a curve. While 20-somethings may win a sprint, performance in many other sports can reach a high later in life. That’s not to mention factors like emotional well-being and mental discipline, which rise and fall in unexpected patterns. And despite nostalgia for the joys of youth, for most of us, our happiest days are actually yet to come.

By learning to recognise these patterns, we can find better ways to nurture our growth and embrace the opportunities available at each stage of life. So what, based on science, are the seven ages of you? And how can you make the most of them?

The seven ages of you

No single stage is the prime of life, as each decade brings new strengths. The trick is to identify them, says **David Robson**

CHILDHOOD Original thinking

It is a great shame we can’t remember our first few years. In terms of the sheer number of changes to the body and brain, early childhood sees the greatest transformations of our lives. We not only learn essential skills for survival – how to walk and feed ourselves – but also language and how to recognise what others are thinking and feeling.

Neurologically speaking, a lot of this transformation involves the steady strengthening of connections between certain brain cells and the pruning of unnecessary connections between others. For some areas, such as the visual or auditory system, this happens rapidly during the first few years. This could explain why childhood is a peak period for learning, especially for sensory skills such as developing the accent of a language or perfect pitch in music. For other brain areas, such as the prefrontal cortex involved in higher level thinking and decision-making, this neural pruning and strengthening continues beyond our teens.

Much of this childhood brain development may arise from a form of statistical learning that resembles the scientific method: making predictions about the world and updating them according to evidence gained through experience. To gather this information, a baby’s attention will drift to anything that is unexpected or surprising – explaining why they are so intensely curious about even the most trivial details. Over time, the process helps them to recognise objects and sounds and to work out what different words mean.

Imaginative play can aid this process, particularly as the child begins to explore the sophisticated thinking that defines our species. Humans engage in counterfactual reasoning, for example, which involves considering complex hypothetical scenarios and exploring the consequences. Playing pretend seems to train that capacity. As developmental psychologist Alison Gopnik points out in her book *The Philosophical Baby*, children spend a huge amount of time in imaginary worlds honing those skills, compared with adults. ➤



MATT MURPHY

This might explain why childhood is a key period for creativity and imagination, with youngsters scoring highly compared with older people on tests of original thinking – thinking up unexpected uses for an object such as a brick, for instance (adolescents generally score more highly too).

As a child grasps more words, a growing ability to tell stories will also affect their ability to remember their life; our autobiographical memory seems to grow with our language skills, which may explain why our recollections of the first few years are hazy at best.

ADOLESCENCE

Risks and rewards

We may think that the wayward teen is a modern invention, but the stereotype can be traced at least to the ancient Greeks. The youth, according to Aristotle, are prone to “overdo everything”. Shakespeare took a similarly dim view: “I would there were no age between ten and three-and-twenty... for there is nothing in the between but getting wenches with child, wronging the ancients, stealing, fighting”.

Puberty – with all those sex hormones rushing through the veins – might seem to be the most obvious reason for this unruly, impulsive behaviour. Until recently, teens were also thought to undergo some characteristic brain changes that impair their capacity to act rationally. The brain’s limbic system, which governs motivation and reward, matures much more quickly than the prefrontal cortex, which is essential for behavioural inhibition and logical thought. As a result, teenagers were thought to have “imbalanced” brains wired to experience uncontrollable emotions, with little capacity to rein them in until their mid-20s, which is when the prefrontal cortex finally catches up with the limbic system. Until that point, adolescents were thought to be incapable of making good decisions – an idea that is still popular today.

In Romer’s opinion, it is time to ditch these stereotypes. “They are a very gross generalisation,” he says. There is actually limited evidence that most teens have a severe deficit of self-control, he says.

It is true that the capacity for “sensation seeking” behaviour – the desire for varied, new and intense experiences – peaks between the ages of 16 and 19, which may explain the willingness of teenagers to take risks. But Romer thinks that scientists should focus more on the many benefits of teen spontaneity and curiosity when trying to explain their risk-taking behaviours. “Adolescents are exploring and trying things out,” he says. “That’s going to involve a certain amount of risk. But you have to try things out in order to learn if they’re successful and adaptive.”

Whether it is their attempts to explore their sexuality or a desire to travel, the drive to seek new sensations helps adolescents to amass a wealth of experience that they can draw on in later life. This is aided by the underappreciated trait called tolerance of ambiguity. Adolescents are particularly good at coping with uncertain outcomes, which is why they are able to embrace new situations so readily.

We also need to appreciate the need adolescents have to establish themselves socially. A stable social network is essential for our well-being as adults. According to some researchers, this could explain why teens are so keen to avoid rejection and are

prone to peer pressure, even if it involves acting recklessly. They may simply calculate that the risks are worth it given the possibility of cementing relationships, which isn’t necessarily an irrational decision if your goal is to set up a secure friendship circle.

TWENTIES

Faster and happier?

For many people, the third decade is the most exhilarating period of their lives, when they launch into the wider world, often establishing a career and meeting their life partner. Little wonder that, looking back from old age, most people have much richer recollections of their early 20s, compared with any other decade – a phenomenon known as the reminiscence bump.

Interestingly, the memories within the reminiscence bump are almost universally positive. Perhaps because it makes a more satisfying narrative of this defining decade, we preferentially recall happier events, while the stresses tend to be forgotten. In reality, you are less happy in your 20s than in adolescence or old age. Nostalgia is often harmless, but it can be helpful to take off those rose-tinted spectacles and question some assumptions about these heady years. It is common, after all, to assume your 20s were your physical and mental prime, suggesting that the rest of your life is an inevitable decline. But the truth is more complicated.

Consider your fitness. It is true that elite swimmers usually reach peak performance at 20, and sprinters tend to do the same at 24 to 26 – after which there is often a steady decline in performance in these sports. This is the result of biological changes, such as the loss of some “fast twitch” muscle fibres, which create the sudden bursts of energy necessary for high speed and explosive strength.

For professional sprinters, this soon creates an insurmountable barrier. “At that level, even a 0.5 per cent decline in overall performance can hold you back,” says Gennaro Boccia at the University of Turin in Italy, who has recently studied age-related changes in the performance of Italy’s top athletes. But such



A teenager’s desire to try new things pays off in later life



Many people think of their 20s as the best time of their life

impacts are generally irrelevant for the rest of us in our everyday attempts to remain fit and active. “In the general population, you only start seeing a decline in your performance after 40,” Boccia estimates.

The brain’s trajectory after 20 is similarly complex and doesn’t represent a simple decline. In a series of experiments, Laura Germine at Harvard Medical School has tested tens of thousands of people to examine the differences in cognitive abilities between age groups. Her findings confirm that 20-somethings do seem to have the edge in measures of reaction time and capacity to solve novel problems quickly. Yet many important skills – including working memory

“Many important cognitive skills reach their zenith in later life”

capacity, face recognition, emotion perception and the ability to sustain concentration – reach their zenith much later in life.

THIRTIES Staying power

If your 20s were defined by speed – both physical and psychological – then your 30s might be considered the decade of endurance.

This is evident in the peak performance of long-distance athletes. For male marathon runners, the ideal age appears to be 31 years old. For women, it is slightly less, 27 years, although champions like Paula Radcliffe have continued to win until their mid-30s. The peak age for a 100-mile ultramarathon, meanwhile, is 37 for men and 38 for women.

Why would this be? The loss of fast-twitch muscle fibres will have little impact for sports that rely on stamina. But people in their 30s do face a drop in aerobic capacity – the body’s efficiency at delivering oxygen to the muscles – which could reduce performance. The extra years of experience may, however, bring the advantage of improved emotional regulation and planning, and these can help athletes to pace themselves during endurance events and to cope with the inevitable stress and exhaustion. This could offset the early stages of the physiological decline, creating a sweet spot in an athlete’s 30s.

The brain is also hitting its stride on an important measure of cognitive ability. Germine has found that performance on certain working memory tasks – such as the capacity to hold multiple pieces of information in mind at once – peaks in our early 30s. It isn’t hard to see how this might be beneficial, as demands of home and work start to build over the decade.

Sure, by this age our mental processing speed is a bit slower, but this loss is a small trade-off for the many other abilities we acquire as we age. “You may not be as fast as you were when you were 20, but you don’t need that [speed] when you are doing the things that you have already specialised in,” says Germine.

FORTIES

Pay attention!

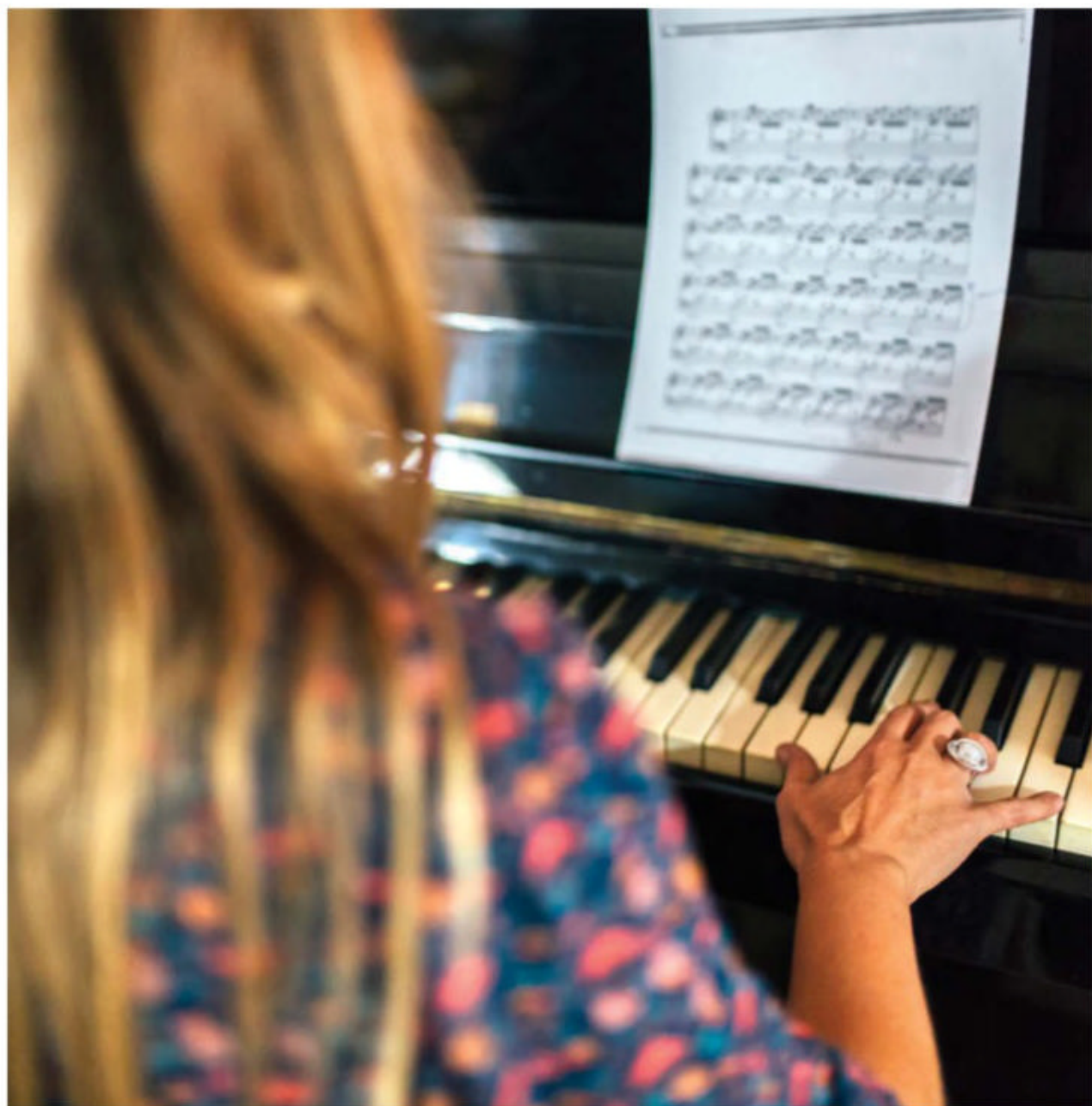
“The really frightening thing about middle age,” the actor and singer Doris Day is said to have quipped, “is that you know you’ll grow out of it.” We may gallantly try to claim that life begins at 40 – but for many people, it can feel more like the beginning of the end.

Midlife wasn’t always seen this way. In Renaissance paintings depicting the stages of life, you often see the decade symbolised as a lion, a sign of courage and strength. It isn’t clear why we have a more negative view today, but Margie Lachman, director of the lifespan development lab at Brandeis University in Massachusetts, suggests it may be linked to the pressures that begin piling up in our 30s. “Midlife is a period of high stress today, more so than in the past,” she says. “One is literally in the middle of work and family careers. This can take a toll on one’s ability to focus on one’s own well-being.”

There are, however, many reasons to feel positive about this pivotal period. Germine’s studies have included the famous “mind in the eyes” test, for instance, which gets people to infer emotional states from small differences in facial expressions. She found that people in their late 40s scored highest. This may be due to practice, she suggests. “When you think about the amount of social nuance that one has to learn across the lifespan – that’s where we think that comes from.”

Germine found similar patterns in a task demanding sustained attention. In this, the participants had to watch different scenes fade into one another and adapt their response according to what they saw – pressing a space bar when they saw a city and releasing it when they saw a mountain. Despite the effort (and potential tedium) of the task, 40-somethings found it much easier to “get into the zone” than younger people.

It is interesting to note that middle-aged people frequently bring in the most supplies in traditional hunter-gatherer societies. According to various anthropological studies, hunter-gatherers often take decades to learn their skills, and these abilities continue to grow into their 40s.



PLAINPICTURE/MARCO GOVEL

Continuing to master new skills throughout life boosts brain health

There are some downsides to hitting this age, of course. Our skin tends to lose some of its elasticity and our body fat starts to be redistributed around the midriff. But after a dip in life satisfaction, happiness is already set to rise at the end of this decade and the beginning of the next.

Contrary to popular opinion, humans seem to have evolved to flourish into middle age and beyond.

FIFTIES AND SIXTIES

Crystallising intelligence

Unless you are extremely lucky, your body will have started to slow down by your 50s and 60s. But that is no reason to stop caring for your health. A growing number of studies show that our psychological outlook and physical lifestyle continue to have enormous consequences in later life. “To a large extent, the way one ages is in one’s own hands,” says Lachman.

Scientists didn’t always promote the benefits of exercise in later life. With the assumption that this was a time of inevitable decay, people

were generally encouraged to take it easy. “We used to think vigorous exercise would be dangerous for older adults, that they might suffer a heart attack or fall or break bones,” says Lachman.

Lachman’s own research has helped to change these views. In the middle of the 1990s, her team began following more than 3000 people aged between 32 and 84. Over the course of a decade, the participants’ general health was measured as well as three potentially protective factors: their physical activity, their social support and their sense of control over their life.

In terms of overall health, Lachman found that those in their 50s and 60s who scored well on those three factors looked much more like those who were in their 30s to 40s in the study than people of their own age.

The potential for interventions is obvious. “Promoting group exercise or sharing one’s exercise successes with friends and family can be a way to increase activity and social support, both of which are beneficial for health,” says Lachman. Talking therapies, meanwhile, might help to change people’s sense of control, encouraging them to see the potential to make positive change in their lives.

We can be similarly proactive about our cognitive functioning. According to Germine’s studies, measures of “crystallised intelligence” – the knowledge, facts and skills that we accumulate through life, such as vocabulary size – peak in our 50s and 60s. That should give you greater expressive power than those in their 20s or 30s. This accumulation of knowledge might also be responsible for some of the reduced processing speed of older people measured in cognitive tests. After all, when recalling information, it takes longer when you have more information to sift through.

You may, of course, encounter the odd moments of forgetfulness. But research by Dayna Touron at the University of North Carolina, Greensboro, suggests that older people are overly pessimistic about the state of their memory, which can needlessly discourage them from exercising their minds. When driving, for instance, they may use a GPS for fear of forgetting directions; yet they often

“To a large extent, the way you age is in your own hands”

can remember the correct route when pressed.

This habit, called memory avoidance, could speed decline, so it is important not to let pessimistic judgements become a self-fulfilling prophecy. Fortunately, there is now plenty of evidence that people who continue to learn new, challenging skills tend to maintain healthier brains in later life. This could be learning a language or musical instrument or a craft like quilting – anything that is complex enough to tax your mind.

With our brains as well as our bodies, it really is a case of use it or lose it.



It is never too late to reap the benefits of getting stronger

SEVENTY-PLUS The older the...

If you want to remain healthy, then regular, challenging activity is essential into your 70s and beyond. “It is never too late to make some changes,” says Lachman. One study, for instance, found that a programme of strength training improved the mobility of people in their 90s. At the same time, you might come to appreciate the wisdom that has accrued during your life and try to put it to good use.

This may sound like a cliché, but Igor Grossmann at the University of Waterloo in Canada, has designed tests that measure various elements of “wise reasoning”, and the age-related changes are revealing. In a typical test, participants are presented with a text describing a conflict – either personal or political – and asked to discuss the potential outcomes. Their answers are then scored on qualities such as intellectual humility (the capacity to admit what we don’t know), the ability to adopt many perspectives and the ability to find a compromise. Together, these traits are believed to capture the general concept of wisdom that has been promoted by philosophers throughout the ages.

Grossmann has found that people’s wise reasoning scores are often more strongly linked with various measures of life satisfaction and the quality of people’s social relationships, than traditional measures of cognitive ability like IQ. And older people seem to ace these tests compared with younger or middle-aged participants. The overall quality of our decision-making really does seem to increase steadily throughout life.

Our impressive abilities at all seven ages of life make it clear that there simply is no single prime period: every decade could be considered a golden age in one form or another. From our entrance into this world to our exit, humans have great potential. ■



David Robson is author of *The Intelligence Trap: Revolutionise your thinking and make wiser decisions*. To buy a copy, go to shop.newscientist.com/the-intelligence-trap

Consciousness expanded

How a physical brain creates the feeling of being is one of life's greatest mysteries – but one we are starting to crack. Join us on a journey into the recesses of human consciousness, guided by the answers to 10 questions that reveal how our everyday experience of reality is more mind-blowing than we imagine

1

What is consciousness?

IN ESSENCE, consciousness is any kind of subjective experience. Being in pain; smelling onions frying; feeling humiliated; recognising a friend in the crowd; reflecting that you are wiser than you were last year – all of these are examples of conscious experiences. In a field fraught with disagreements, this is something that most, but not all, researchers agree on. Go any deeper, though, and the rifts open up.

The 17th-century French philosopher René Descartes famously divided the universe into “matter stuff”, such as rocks and physical bodies, and “mind stuff”. In the 20th century, philosopher David Chalmers at New York University built on Descartes's separation, known as “dualism”, and the work of later thinkers, to distinguish between “easy problems of consciousness” and “the hard problem”.

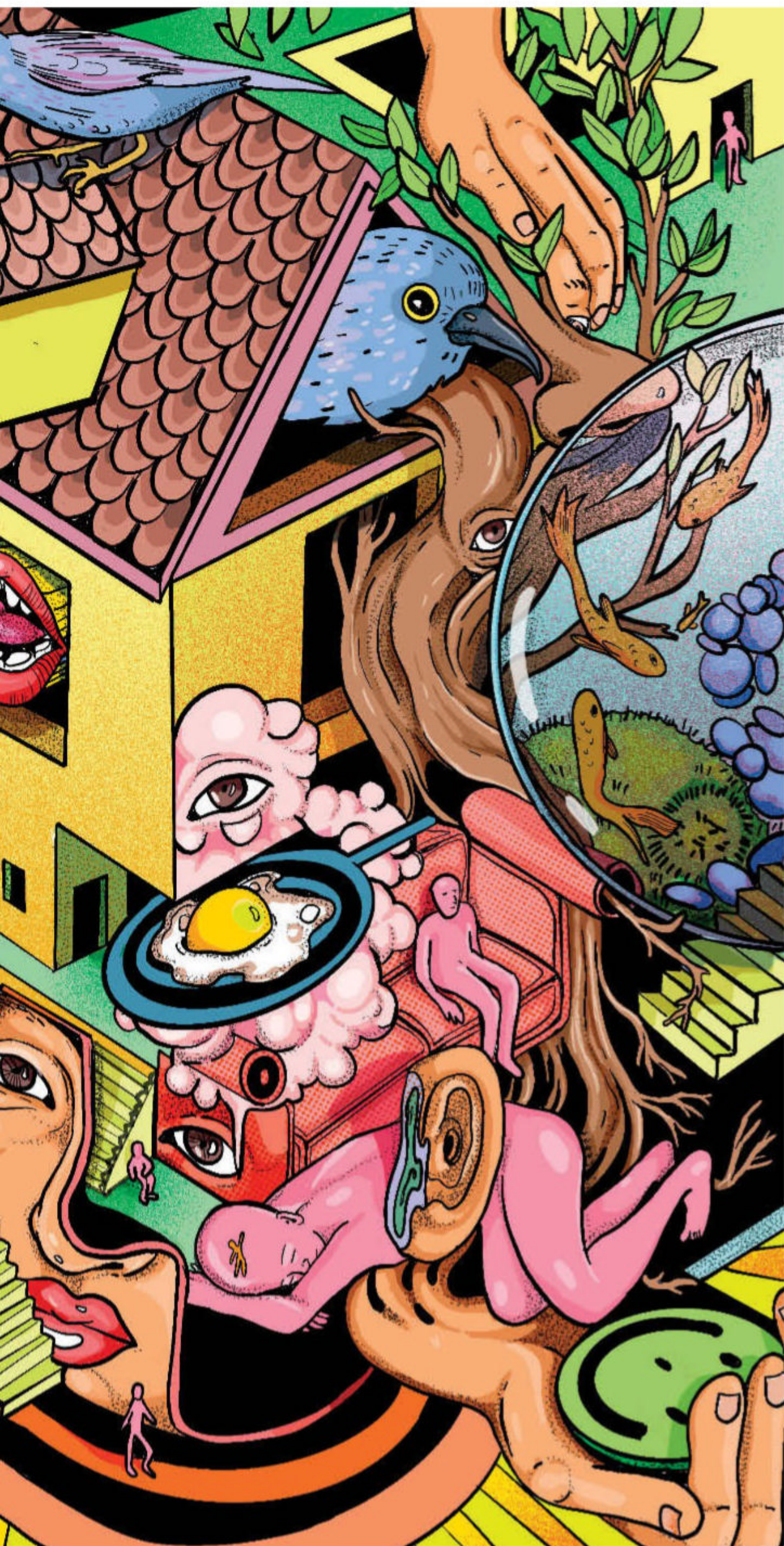
The easy stuff consists of explaining the brain processes associated with consciousness, such as the integration of sensory information, learning, thinking and being awake or asleep. Though we are making steady progress, these problems have yet to be cracked: they are easy

only in the sense that the known strategies of cognitive and neuroscientific research should eventually provide full explanations.

The hard problem, which Chalmers introduced at a scientific meeting in 1994, is to explain why and how we have subjective experiences at all. “Consciousness poses the most baffling problem in the science of the mind,” Chalmers said. When we think and perceive, there is a “whirl of information-processing” in the brain, as he put it, but also very distinctive subjective states of mind. The puzzle is how a 1.3 kilogram organ with the consistency of tofu can generate the feeling of being.

Some believe the hard problem can never be solved – it is beyond the capabilities of human cognition. Others think that, with a better understanding of the brain, it will simply go away. Anil Seth at the University of Sussex, UK, sees a third way. He thinks that tackling the easier stuff will allow researchers to address the hard problem indirectly. To this end, he has developed a framework that highlights different aspects of consciousness to be





EVA REDAMONTI

investigated. Other theorists divide it up differently, but for Seth the three main components are conscious content, conscious self and conscious level.

Conscious content is whatever you are conscious of, including awareness of sensory perceptions. Conscious self is a unique component of conscious content that refers to self-awareness – the subjective feeling of being you – and includes being aware of your own awareness and reflecting on your conscious thoughts, known as meta-consciousness. And conscious level refers to differences in so-called “global states”, such as being asleep, under anaesthesia or being awake (see “How many states of consciousness do humans have?”, page 36).

These aspects of consciousness aren’t independent, says Seth. “If you explain all the different kinds of conscious content – everything you might be conscious of – then you’ve probably explained everything,” he says. But he thinks breaking consciousness down like this is the best approach to tackling what it is. **Emma Young**

2

How many states of consciousness do humans have?

Consciousness was traditionally thought to be like a light switch: it is on when you are awake and off when you are sleeping, anaesthetised or in a coma. However, when we dream while asleep we often have subjective experiences that are similar to our waking ones. Taking this into account led to the conclusion that, instead of two states of consciousness, there are three.

This idea didn't hold for long, though. One driver of the shift in thinking was research looking at people who are in a coma, where the brain is considered to be entirely unresponsive, and others who are in a persistent vegetative state, which is marked by a sleep-wake cycle, but with no response to stimuli. People in the latter state would appear to be more conscious than those in a coma, and both seem less conscious than someone under mild sedation. Neuroscientists and philosophers alike began arguing that there are many states of consciousness: it is more like a dimmer-switch or a ladder with zero consciousness at the bottom and maximum consciousness at the top.

This picture was further complicated by a startling finding: a few people diagnosed as being in a persistent vegetative state have brain activity indicative of some awareness. In 2010, for example, a team that included Adrian Owen, then at the University of Cambridge, reported several cases of people who could respond to questions using their brain activity, picked up by an fMRI brain scanner. These included a 29-year-old man who was able to give yes or no answers to queries by imagining doing a different activity for each word.

Next came claims of levels of consciousness beyond alert

wakefulness. In 2017, a team led by Anil Seth at the University of Sussex, UK, reported that people who had taken LSD or psilocybin (the hallucinogenic compound in magic mushrooms) experienced a "higher" level of consciousness: at least, their brains showed more diverse electrical activity with the drugs than without them. The ladder of consciousness was expanded to accommodate this and other states, including hypnosis, sleepwalking, epilepsy-induced unawareness, being half asleep (sleep is no longer regarded as an all-or-nothing phenomenon) and daydreaming.

Multi-faceted state

Now, the ladder model itself is being challenged. Owen is in the vanguard. He and two colleagues have described it as "untenable". They argue that, just as someone who is blind is no less conscious than a sighted person, so someone who has taken a mild sedative, say, is no less conscious than someone in a regular state of wakefulness.

Think about a toddler's experience of the world, says Owen. An adult might look at the Eiffel Tower and be aware of its name and location and what was happening in their life when they last saw it, whereas a toddler might be conscious of just a big tower. In some ways, but not all, their conscious experience of the world isn't the same as an adult's. If that adult developed Alzheimer's disease, they may then come to have a similar awareness of the Eiffel Tower as a toddler, but their conscious experiences wouldn't be the same. "Consciousness is multi-faceted," says Owen, "and trying to measure it along any single dimension, including 'depth', will inevitably be futile."

Instead, he and his colleagues propose a more complex, multidimensional framework. Rather than being points on a line, states of consciousness would be represented by a spider's-web-type shape, with spikes indicating the extent to which someone possesses each dimension. Unfortunately, we don't yet know enough about consciousness to say what many of these dimensions are, let alone measure them. That is a concern to researchers, including Seth. Nevertheless, the idea has been influential. "It's on the money," says Seth. Emma Young

There is an ongoing debate about degrees of consciousness



TOMMY TRENCARD/ALAMY

3

Can physics explain consciousness?

“Consciousness may emerge from quantum collapse in the brain”



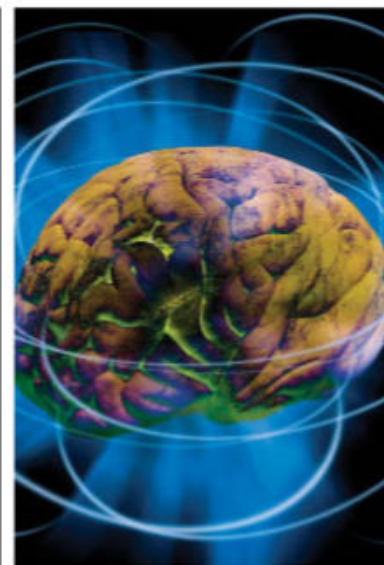
If physics explains all the phenomena in the universe, and if consciousness is part of the universe, then it seems that physics can explain consciousness.

Of course, this assumes that consciousness isn't separate from the material reality that physics explains – which runs counter to René Descartes's dualist view of mind and matter. Some have no problem with that. They include Daniel Dennett at Tufts University in Massachusetts and Michael Graziano at Princeton University, who argue that our intuitive sense that consciousness needs an explanation that goes beyond objective descriptions of the physical world is misplaced. Consciousness is a mirage produced by sophisticated neural mechanisms in the brain, they contend, so we need no new physics to explain it. Rather, we need a better understanding of how the brain creates models: of the world, of a self in the world and of a self subjectively experiencing the world.

Other non-dualists don't outright deny that consciousness may have unusual properties that need explaining. If they are correct, then quantum mechanics may offer an explanation.

Quantum systems can exist in a superposition of all possible states simultaneously, and classical reality emerges when this superposition collapses into a single state. One idea is that this happens when the mass of a quantum system crosses a threshold. According to theoretical physicist Roger Penrose at the University of Oxford and anaesthesiologist Stuart Hameroff at the University of Arizona, consciousness emerges as the result of such collapses occurring in the brain. In their model, called orchestrated objective reduction (Orch OR), microscopic structural elements within neurons, called microtubules, enter into states of quantum superposition. These span networks of neurons and when the mass of the microtubules in a superposition exceeds some threshold, it collapses, producing conscious moments.

There are many dots to be connected for this idea to be taken seriously. One is to show that microtubules can attain states of quantum superposition in the relatively balmy



If physical processes in a brain create consciousness, what are they?

VICTOR DESCHWANBERG/SCIENCE PHOTO LIBRARY

temperatures of the human brain. According to Hameroff, preliminary and unpublished work by Jack Tuszynski at the University of Alberta, Canada, and Gregory Scholes at Princeton University has achieved that. “They have found evidence for quantum states in microtubules persisting for up to 5 nanoseconds,” he says.

The next step is to expose microtubules to anaesthetics, which selectively disrupt consciousness while leaving non-conscious brain functions intact. “The prediction is that anaesthetics would dampen microtubule quantum states proportional to known anaesthetic potencies in putting humans and animals to sleep,” says Hameroff. This provides a way to test Orch OR, he says, and Bruce MacIver at Stanford University in California is looking into conducting such an experiment.

But what if consciousness is separate from material stuff and so outside the purview of physics? If so, Orch OR probably has it backwards, says Johannes Kleiner at the Munich Centre for Mathematical Philosophy in Germany.

In a yet to be published paper, he and Kobi Kremnitzer at the University of Oxford show mathematically that if non-material consciousness were to affect material reality, in our physical reality, superpositions would appear to spontaneously collapse. “So the arrow is from consciousness to collapse,” says Kleiner. “In Orch OR, it's from collapse to consciousness.” **Anil Ananthaswamy**

4

What is consciousness like in other animals?

Children know the fun of throwing a ball into the sea, only to watch the waves fling it back. Jennifer Mather and Roland Anderson at the Seattle Aquarium were surprised to find octopuses playing similar games. Their toy was a floating pill bottle, which they were free to ignore or explore as they wished. Six of the aquarium's octopuses soon lost interest, but two showed childlike curiosity, pushing it with their arms or shooting jets of water to move it against the tank's current. It is hard to interpret this as anything other than play, which many researchers argue requires some form of conscious awareness.

Many animals exhibit behaviours similarly suggestive of an inner life. Conscious creatures may include our primate cousins, cetaceans and corvids – and potentially many invertebrates, including bees, spiders and cephalopods such as octopuses, cuttlefish and squid. The challenge, of course, is to understand how the inner lives of these creatures differ from our own.

In the past, scientists spoke about “levels of consciousness”, as if there were a hierarchy with humans on top. But in a paper published in 2020, Jonathan Birch, a philosopher at the London School of Economics and his colleagues argue that we would do better to consider five separate elements of conscious experience. The first is perceptual richness, or how well an animal can discriminate different details in each of its senses. The second is evaluative richness – broadly speaking, the capacity to differentiate between positive rewards and noxious stimuli, which could be analogous to human emotions such as pleasure or pain. The third, unity, concerns the extent



Each arm of an octopus could be independently conscious

AZOR WILDLIFE PHOTO/ALAMY

to which an animal integrates the information from its sensory organs into a single experience. Then there is temporality: does the creature's past experience influence its present behaviour, and can it plan for the future? The fifth dimension is selfhood. This may be tested by assessing whether an animal recognises itself in the mirror, or has so-called theory of mind – the ability to understand that another animal has its own mind.

Many ways of being

According to Birch and his colleagues, it doesn't make sense to ask whether one animal is more or less conscious than another, since each species may score highly on some of these facets of consciousness, but low on others.

Research by Birch's co-author Nicola Clayton at the University of Cambridge offers a good example.

Scrub jays, a type of corvid, bury food to eat later, which demonstrates high temporality because it involves planning for future scarcity and remembering the location of caches. Their use of deception when hiding food in the presence of a rival bird, meanwhile, shows theory of mind, which suggests a relatively sophisticated sense of selfhood, too.

Cephalopods, by contrast, haven't yet shown evidence of self recognition. But the octopuses' enjoyment of play may be a sign that cephalopods can experience something akin to pleasure – evidence of some evaluative richness. They also have extraordinary perceptual richness, with complex vision that can detect polarised light and the ability to taste-by-touch with their suckers.

Perhaps the most startling difference between species concerns the unity of their conscious

experience. We have two eyes but we seamlessly integrate the two visual fields into a single conscious experience, thanks to the thick nerve tract connecting our left and right brain hemispheres. Birds lack that connective structure, leading Birch and his colleagues to speculate that within each individual there may be “a pair of conscious subjects, intimately cooperating with each other”.

Or consider the octopus. Two-thirds of its neurons are located in its arms, and there is some evidence that each limb operates semi-autonomously. “You could conceive of there being eight conscious experiences associated with the different arms, that are partially unified with the experience associated with the brain,” says Birch.

That is a consciousness so alien, it is almost impossible for us to imagine. David Robson

5

When did consciousness evolve?

Given that there is a variety of conscious experience across the animal kingdom, it is natural to ask when and how consciousness first emerged. Can we thank a single common ancestor? Or did it evolve independently in different lines?

Andreas Nieder at the University of Tübingen in Germany favours the latter explanation. He recently conducted an ingenious experiment to discover the brain processing underpinning visual consciousness in corvids such as crows. The birds were trained to respond to different coloured squares, some of them almost imperceptibly faint. Neurons in a region called the pallial endbrain lit up whenever the crows reported seeing the squares, but not when they failed to spot them, suggesting that this area is essential for their conscious visual perception. In humans and primates, a different part of the brain, the prefrontal cortex, performs the same job. For this reason, Nieder thinks that consciousness probably emerged separately on multiple occasions, in much the same way that wings appeared separately in insects, birds and bats.

Eva Jablonka at Tel Aviv University in Israel and Simona Ginsburg at the Open University of Israel take a different view. In their book *The Evolution of the Sensitive Soul*, they argue that birds, mammals and other vertebrates inherited consciousness from a common ancestor. Their argument hinges on a type of thinking known as unlimited associative learning, which involves learning about novel stimuli and linking them together even if time has passed between experiencing two stimuli. It also entails forming chains of such links so that the same stimulus can be interpreted in different ways depending on the context.

Importantly, these associations can be updated: something that triggered danger may come to be a sign of reward if circumstances have changed. Jablonka and Ginsburg believe that this kind of complex learning is impossible without consciousness to direct attention, evaluate different cues in the environment and integrate experiences over time. So, what can it tell us about the origins of consciousness?

Unlimited associative learning has been documented in many diverse species. “Even

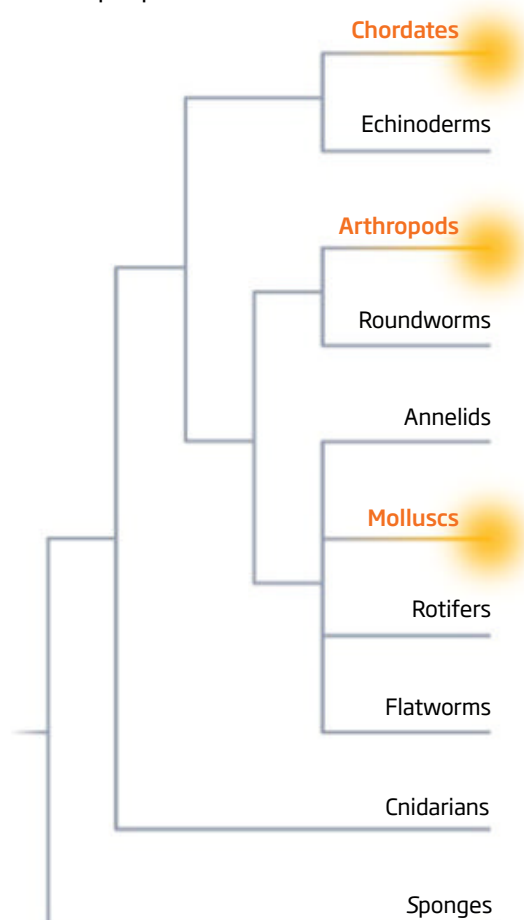
little fish are able to do this,” says Jablonka.

That leads her and Ginsburg to propose that consciousness must have emerged in early vertebrates around 530 million years ago.

They suspect certain groups of invertebrates, including arthropods like honeybees and spiders, can also think in this way. If so, their earliest ancestors evolved consciousness independently of vertebrates, around 500 million years ago. The shell-less cephalopods are relative latecomers to the table. The fact other molluscs don’t appear capable of unlimited associative learning suggests to Jablonka and Ginsburg that consciousness in octopuses, cuttlefish and squid originated around 300 million years ago. “I think it is one of the most promising approaches we have at the moment for how to study the evolution of consciousness,” says Jonathan Birch at the London School of Economics. **David Robson**

The enlightened ones

Signs of consciousness have been found in animals from at least three different **phyla**, suggesting it evolved more than once and is far more common than most people think



Models of consciousness

There are many ideas about how consciousness arises. Here are five of the most influential. Although each has its own proponents, they aren’t all mutually exclusive.

GLOBAL NEURONAL WORKSPACE

Information entering the brain from the outside world or the body competes for attention in the cortex and a structure in the centre of the brain called the thalamus. If the signal it generates is stronger than signals from other information, it is broadcast across the brain into the global workspace. Only then do you consciously register it.

ATTENTION SCHEMA

The brain evolved to contain a model of how it represents itself. This attention schema is like a self-reflecting mirror. It is what creates the subjective feeling of consciousness. There is no “ghost in the machine”; consciousness is just a mirage created by sophisticated neural processing.

PREDICTIVE PROCESSING

The brain is a prediction machine, meaning that what we perceive is the brain’s best guesses about the causes of its sensory input. As a result, much of conscious experience and selfhood is based on what we expect, not what is there.

INTEGRATED INFORMATION

Consciousness isn’t confined to brains. It arises in any system as a result of the way information moves between its subsystems. The degree of integration of this information is measured with a metric called phi. Any system with a phi of more than zero is conscious.

ORCHESTRATED OBJECTIVE REDUCTION

Quantum mechanics can explain consciousness. Microscopic structural elements within the brain, called microtubules, can exist as a superposition of all possible states. This quantum system collapses into a single state when the mass of the microtubules in it exceeds a certain threshold. The collapse is what creates consciousness. **Kate Douglas**

6

How would we know if a machine were conscious?

There is disagreement over whether machines can ever be conscious, let alone how we would know if one were. Your view may depend on how you see consciousness.

If the subjective feeling of consciousness is an illusion created by brain processes, then machines that replicate such processes would be conscious in the way that we are. How would we know this? Daniel Dennett at Tufts University in Massachusetts thinks a Turing test, in which a machine has to convince a human interrogator that it is conscious, should, if conducted “with suitable vigour and aggression and cleverness”, be enough.

Michael Graziano at Princeton University thinks we could take a more direct approach. His attention schema hypothesis sees consciousness as the brain’s simplified model of its own workings – a representation of how it represents things. He believes it is possible to build a machine that possesses a similar self-reflective model. “If we can build it in a way that we [can] see into its guts, then we will know this is a machine that has a rich self-description,” he says. “It is a machine that thinks and believes it has consciousness. And those are confirmable because you can understand, in principle, how the machine is processing information.”

For Graziano, consciousness could appear in any machine, whether it is purely in software or constructed of matter, biological or otherwise. Anil Seth at the University of Sussex, UK, isn’t so sure. “I think it is still an unknown whether consciousness is substrate-independent,” he says. For him, determining whether a machine is conscious requires making informed judgements based on whether, for

example, it has analogues of brain structures that we know are important for consciousness in humans, and what it is made of (brain organoids, for example, are made of biological material).

Identifying consciousness in a machine may be more straightforward if you subscribe to the integrated information theory of consciousness. In principle, this simply entails ensuring that ϕ , a quantity indicating the degree of information integration within the system, is greater than zero. In practice, calculating ϕ is computationally intractable for anything but the simplest of systems. So, even if a machine were designed to integrate information, it would be far beyond our abilities to tell whether it is conscious.

Disintegrated circuits

Phil Maguire at the National University of Ireland, Maynooth, goes further. He notes that, by definition, integrated systems can’t be understood by looking at their parts. “Machines are made up of components that can be analysed independently,” he says. “They are disintegrated. Disintegrated systems can be understood without resorting to the interpretation of consciousness.” In other words, machines can’t be conscious.

Selmer Bringsjord at Rensselaer Polytechnic Institute in Troy, New York, agrees – but for different reasons. He thinks our subjective feeling of being conscious is the outcome of non-material stuff of some sort, and that this is crucial for some of our intelligent behaviour. For him, machines can never possess this essence, so will never be conscious or intelligent in the way that we are. Anil Ananthaswamy

7

What is consciousness for?

Consciousness has deep origins and appears to have evolved several times (see “When did consciousness evolve?”, page 39). This alone suggests that it must provide important benefits when it comes to survival. But what might these be?

Biologists hadn’t grappled with this question until quite recently. Many of those who are now considering it, think one probable answer is flexibility of behaviour. “We are better able to react to the contents [of the conscious mind], as opposed to the unconscious content that will usually trigger automatic reflex-like responses,” says Andreas Nieder at the University of Tübingen, Germany. Being conscious also offers a way to evaluate the world – through feelings that make an individual aware of whether something is good or bad. And consciousness entails selective attention, which allows the brain to focus its processing activity on what is most important.

As a result of all this, instead of reacting in set ways to given stimuli, conscious animals can make complex decisions. “That would be a survival benefit that helps an individual, and then the entire species, to survive better in the changing world,” says Nieder.

Greater behavioural flexibility also fits with the idea that consciousness is associated with a particular type of thinking called unlimited associative learning – the ability to combine multiple cues into a single perception. This allows animals to respond flexibly to the challenges they face, rather than relying on hardwired behaviours. It means, for instance, that they can better discriminate between a healthy and a poisonous source of food based on small perceptual differences. “Associative learning is one of the greatest adaptations on the planet,” says Eva Jablonka at Tel Aviv University in Israel.

According to Jablonka and Simona Ginsburg at the Open University of Israel, consciousness first emerged around 530 million years ago and the resulting behavioural complexity it brought had huge knock-on effects. It created greater competition among species, driving the rapid diversification of animal species known as the Cambrian explosion. Predators, for example, would have been better at

8

Is consciousness detectable in the brain?



eva reda monti

“The emergence of consciousness led to much of the beauty in nature”

detecting their prey, which, in turn, would have needed to find new ways to avoid detection – pushing the predators to become even more sophisticated in their strategies. “There is a kind of ongoing co-evolutionary arms race,” says Jablonka.

But the emergence of consciousness didn’t just allow animals to adopt more complex behaviours. Jablonka suggests that it is responsible for much of the beauty that we see in nature too. It led different species to evolve camouflage, for example. And it pushed plants to evolve colourful flowers that would stick out from the competition to attract pollinating insects. “It changed the world completely,” she says. “The world would have been a very different place, and a very much more boring place, without consciousness.” **David Robson**

One of the most famous outstanding wagers in science is that by 2023, a specific signature of consciousness in the brain will have been identified. Betting against this wager – made in 1998 and with a case of fine wine at stake – is David Chalmers at New York University, the originator of the “hard problem” of consciousness (see “What is consciousness?”, page 34). Betting on it is Christof Koch, who heads the Allen Institute for Brain Science in Seattle, Washington. With two years to go, where do things stand?

We aren’t there yet. Progress has been made in understanding which brain regions and networks are involved, but there is still fierce debate.

Some researchers ascribe a vital role to the prefrontal cortex (PFC), which is responsible for sophisticated cognitive processes including attention, decision-making and planning. They argue that for sensory information to become a conscious perception – for you to actually see a red apple, say – it has to be processed not just by the sensory cortex, but also here. Neuroimaging studies of people and macaques support this idea. But sceptics say that the PFC activity they show could relate to thinking about a stimulus and planning a response, rather than being conscious of it. Koch also points to patients who have had large regions of their PFC surgically removed due to tumours or epileptic seizures. “They go on living, by and large, a normal life, never complaining that they have been turned into zombies,” he says.

To probe further, research published this March reviewed evidence from studies that involved inserting electrodes into various parts of human brains and asking volunteers about the effects. Omri Raccach at New York University and his colleagues concluded that stimulation of only two regions of the PFC, the orbitofrontal cortex and the anterior cingulate cortex, sometimes ➤

alters reports of conscious experience. Based on what is known about the function of these subregions and what the participants reported, it seems likely that they support the emotional aspects of conscious experience, as well as self-consciousness and meta-consciousness (the awareness of being aware). But they may not be involved in more fundamental sensory perceptual awareness. It appears that an area towards the rear of the brain's cortex – the “posterior hot zone”, as Koch calls it – is crucial for this.

Joined-up thinking

One posterior region apparently important for consciousness is the parietal cortex, which processes sensory information from the body. Earlier this year, Mohsen Afrasiabi and Michelle Redinbaugh at the University of Wisconsin-Madison and others reported research on macaques that were sleeping, anaesthetised or awake. They concluded that connectivity between the parietal cortex and two other brain regions, the striatum and the thalamus, is a “hallmark” of conscious states. “Our findings highlight the importance of integration between parietal and subcortical structures and challenge a key role of the frontal cortex in consciousness,” says Afrasiabi.

Studies of people who have regained wakeful consciousness after being in a vegetative state also support the idea that integration of activity in different regions is key for various aspects of consciousness. This notion is central to the global neuronal workspace model too (see “Models of consciousness”, page 39). “While there is much disagreement about which brain areas are most crucial for subjective experience, most theories do agree that consciousness is dependent on interactions and communication within and between different brain areas,” says Redinbaugh. The exact nature of the interactions isn't yet clear as current technology can't measure everything that needs to be measured at once.

With the clock ticking on his wager, Koch anticipates defeat. “The extent to which more frontal regions of the cortex, let alone other brain regions, contribute to consciousness will remain open for many years to come,” he says. “After all, the brain is the most complex piece of active matter in the known universe.” Emma Young

9

Is the universe conscious?



Panpsychists tend to view the universe as conscious

MATTEO VIVIANI/LAMY

For some, the question of whether the universe is conscious makes little sense. “It presupposes a view of consciousness as some special something which is detached from psychology,” says Daniel Dennett at Tufts University in Massachusetts. “Is the universe embarrassed? Is the universe happy? If it can't be any of those things, then the claim that it's conscious is, I think, vacuous.”

Others don't dismiss the question outright, however. In particular, the integrated information theory (IIT) of consciousness raises the spectre that any physical system can be conscious. A metric called phi measures how much integrated information a system possesses, and any with a phi that is even a smidgen more than zero is conscious. That would include, for example, a thermostat, which might simply be conscious of being on or off. Hard-line proponents of IIT are led towards panpsychism, the idea that everything in the universe is conscious – including, possibly, the universe itself.

In February, mathematician Johannes Kleiner at Ludwig Maximilian University of Munich, Germany, and Sean Tull at Cambridge Quantum Computing Limited, UK, came a step closer to a formal analysis of the idea. They published what they call the mathematical structure of IIT. They identified the mathematical space that describes the states of physical systems on the one hand, and the mathematical space that describes experiences that physical systems may be having on the other. They then developed an algorithm to map the elements of one space to the other.

If IIT says a physical system is conscious because its phi is greater than zero, then this new maths lets you map its physical state to an experience.

So, in this way of thinking, is the universe conscious? “The interesting bit is that you can ask that question of the theory,” says Kleiner. “You can plug in the state of the universe, if you happen to be able to describe it, and then the mathematical mapping would give you an answer.” Unfortunately, he adds, right now such calculations are possible for only the trivially simplest of systems.

Philosopher Kelvin McQueen at Chapman University, California, is more circumspect about IIT and its implications for panpsychism. Panpsychism as an idea originated because some philosophers went searching for the intrinsic nature of the material world and came up empty, he says. Take an elementary particle's inertial mass, which is defined as its disposition to resist acceleration given an applied force. “But what is mass in and of itself that gives rise to that disposition? What's its intrinsic nature?” he says. “The panpsychists like to say, ‘well, the only thing that's intrinsic that we know of is consciousness, so mass is ultimately really a kind of consciousness.’ They imbue consciousness in elementary particles.” However, according to IIT, an individual elementary particle isn't a system that is integrating information. So IIT isn't synonymous with panpsychism. “It doesn't say everything is conscious,” says McQueen. And the universe could be one of those things that isn't, even though it has conscious systems within it. **Anil Ananthaswamy**

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Does consciousness make reality?

To answer this question, we must first consider how the reality of everyday experience emerges from the more esoteric quantum reality.

A quantum system exists in a superposition of all possible states, which collapses into a single state (classical reality) only when someone or something observes or measures it – at least, according to standard quantum theory. However, this theory fails to define exactly what constitutes a measurement or an observer, and how exactly that collapse happens. Since the 1950s, many theorists have tried to solve this “measurement problem” by getting rid of the need for collapse-causing measurements and observers. The Many Worlds interpretation of quantum mechanics, for example, says that

there is no collapse and that each classical state in the superposition is physically realised in a different world. So-called collapse theories, meanwhile, propose that superpositions collapse randomly into one of the many possible classical states once the mass of the matter in the quantum system crosses some threshold, and so resists superposition. There is no empirical evidence to prove or disprove these ideas.

Another possibility is that consciousness causes collapse. The Nobel prizewinning theorist Eugene Wigner speculated just this in the 1960s. He eventually disavowed his own meanderings, however, and the idea went out of favour, partly because there was no way to formulate such a theory with

mathematical precision. Recently, however, David Chalmers at New York University and Kelvin McQueen at Chapman University in California came up with an admittedly speculative but mathematically grounded account of how consciousness can cause collapse.

Quantum collapse

They start with integrated information theory (IIT), taking it as an exemplar of a mathematical account of consciousness. IIT says that any system that integrates information is conscious. Chalmers and McQueen consider IIT as applied to quantum systems, reasoning that any such system that integrates information can enter a superposition of conscious states.

They then posit that conscious states are resistant to superposition, in much the same way that, in other models, mass that crosses some threshold resists superposition and causes collapse. So if a quantum system enters a superposition of states in which at least one of the states is conscious (according to IIT), then that consciousness will collapse the system.

In this way of thinking, “consciousness creates classical reality”, says McQueen. “But it doesn’t create quantum reality. It’s converting quantum reality into classical reality.”

Previous attempts to solve the measurement problem by appealing to consciousness have run into a major problem: if classical reality requires the presence of conscious humans, how did the universe evolve classically to the point where human consciousness appeared? The new idea avoids this because IIT doesn’t limit consciousness to biological beings. The universe could have begun as a quantum system and continued evolving quantum mechanically until matter first became able to integrate information. This consciousness then started to collapse quantum reality, creating the sort of classical reality we experience today.

Anil Ananthaswamy ■

Emma Young’s new book is *Super Senses: The science of your 32 senses and how to use them* (John Murray)

Anil Ananthaswamy’s latest book is *Through Two Doors At Once* (Dutton)

David Robson is the author of *The Intelligence Trap: Revolutionise your thinking and make wiser decisions* (Hodder & Stoughton)



If you don't look, does classical reality actually exist?

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